

Week 06 Reading: Nonstationary flood frequency at Brays Bayou (bring two questions Wednesday)

What changes in a flood frequency analysis when the watershed changes

2026-09-28

Read the whole paper before Wednesday's class (9/30), and bring two questions to class: one factual question about something in the paper you did not follow, and one bigger-picture question about what the analysis means or where it could go wrong.

Assigned

1. Smith et al. (2026) (open access)

You should be able to follow nearly all steps of this paper. One thing to note is the authors fit a log-Pearson Type III (LPIII) distribution as mandated by US federal flood frequency guidelines (Bulletin 17C) require. For this reading, treat it as a lot like the GEV: a three-parameter distribution with a location, a scale, and a shape parameter that controls how heavy the upper tail is, fitted here to the logarithm of the annual peak flows.

The Bayesian estimation, the posterior predictive frequency curves, and the credible intervals are the same ideas you used in labs 4 and 5.

Questions to bring

1. **Factual.** A question about something specific in the paper: a method, a figure, a number, or a modeling choice you could not follow.
2. **Bigger picture.** A question about what the results mean, whether you would trust them for a decision, or what the analysis leaves out.

Bibliography

Smith, C. H., Skahill, B., & Margo, D. A. (2026). Nonstationary Flood Frequency Analysis for Urban Watersheds Using Open-Source Bayesian Software: Contrasting Case Studies from Texas. *Water*, 18(5), 636. <https://doi.org/10.3390/w18050636>